Mohua Akter

GitHub Portfolio LinkedIn
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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate specializing in data science and machine learning, with hands-on experience across RAG, LLMs, computer vision, and applied statistics. Designed and evaluated multilingual chatbot architectures with strong semantic-similarity performance across Bangla, English, and Banglish; benchmarked semantic segmentation models to 0.944 mIoU. Seeking entry-level data science, machine learning, or ML research roles.

HONORS AND AWARDS

Fully Funded Merit Scholarship — East West University **Summer 2025–Spring 2026**
Dean's Scholarship — East West University Fall 2024–Summer 2025
Cum Laude — East West University
Academic Honors: Dean's List (Fall 2024, Fall 2025)

EDUCATION

East West University — Dhaka, Bangladesh *October 2022 – August 2026*
BSc in Computer Science and Engineering
CGPA: 3.76/4.00 | 140 credits earned
Relevant Coursework: Machine Learning, Artificial Intelligence, Big Data Analytics, Statistics for Data Science, Statistics and Probability, Data Structures, Algorithms, Database Systems

TECHNICAL SKILLS

Machine Learning / AI: PyTorch, RAG, FAISS, ChromaDB, LLMs (Gemma, Qwen, Phi-2), self-supervised learning, semantic segmentation, InsightFace, ArcFace-ResNet50

Data & Databases: SQL, MySQL, Supabase, SQLite, Big Data Analytics, Statistical Analysis

Programming: Python, C, C++

Web: HTML, CSS, JavaScript, PHP, Flask

Tools and Platforms: Git, GitHub, Docker, Microsoft Excel, PowerPoint, Word, Google Sheets, Google Docs

Languages: Bangla (native), English (fluent)

MAJOR PROJECTS

Multilingual University Chatbot — Capstone GitHub
Designed the RAG pipeline and hybrid RAG+LLM chatbot, moving from Gemma 3 to Qwen; evaluated five Rasa, RAG+LLM, and hybrid architectures for Bangla, English, and Banglish in low-resource settings. Best architecture achieved the strongest semantic-similarity (BERTScore) results across all five designs. Supervised by Dr. Mohammad Rezwanul Huq, EWU.

FruitSeg30 Segmentation Benchmark — Solo GitHub
Benchmarked DeepLabV3, SegFormer, and YOLOv26 on 1,969 images across 30 fruit classes; YOLOv26 achieved 0.944 mIoU, outperforming both alternatives. Published processed train/validation/test splits as public Kaggle datasets.

EWU Admission Chatbot — Solo GitHub
Built a RAG admission chatbot and fine-tuned Gemma 3 on custom JSONL data covering EWU programs, tuition, policies, and the admissions calendar. *Tech: Gemma 3, RAG, LLM fine-tuning.*

AttendX — Smart Attendance System, Solo GitHub
Built a web attendance system with real-time face recognition and role-based student, teacher, and administrator portals. *Tech: InsightFace, ArcFace-ResNet50, Flask, Supabase, SQLite.*

Learnova — LMS / Student Engagement Platform, Solo GitHub
Built an AI-powered learning platform with attendance tracking, classroom management, and analytics; deployed on Vercel. *Tech: PHP, XAMPP, MySQL, HTML, CSS, JavaScript.*

Paris2024 Database Management System GitHub
Designed and implemented a relational database system for managing Olympic event data. *Tech: SQL, MySQL.*

University Multi-Campus Network Design — Solo GitHub
Designed a multi-campus network topology and addressing scheme for a university setting.

RESEARCH INTERESTS

Data Science, Machine Learning, Deep Learning, Natural Language Processing, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Computer Vision, Semantic Segmentation, Self-Supervised Learning, Applied Statistics, Big Data Analytics, Low-Resource and Multilingual NLP.

ONGOING RESEARCH

Ninapro DB7 EMG Self-Supervised Learning *In progress*
Testing and comparing self-supervised learning methods on the Ninapro DB7 EMG dataset to identify the most effective approach. *Tech: Python, PyTorch, SSL.*

Smart Contract Rug-Pull Detection *In progress*
Reviewing research on rug-pull detection in smart contracts; developing an approach centered on intent and behavior-deviation analysis.

EXPERIENCE

Private Tutor — Self-employed, Dhaka, Bangladesh *2021–Present*

- Sole income source while completing a full-time Computer Science and Engineering degree with a 3.76 CGPA.
- Teach Physics, Chemistry, Higher Mathematics, General Mathematics and Biology to Grades 9–12.

LEADERSHIP AND ACTIVITIES

Event Organizer – Jatrabari Ideal School and College
Anchor/Host – Jatrabari Ideal School and College

REFERENCES

Dr. Mohammad Rezwanul Huq
Professor, Department of Computer Science & Engineering
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